



**EDU TERIA**

**72<sup>nd</sup> BPSC**

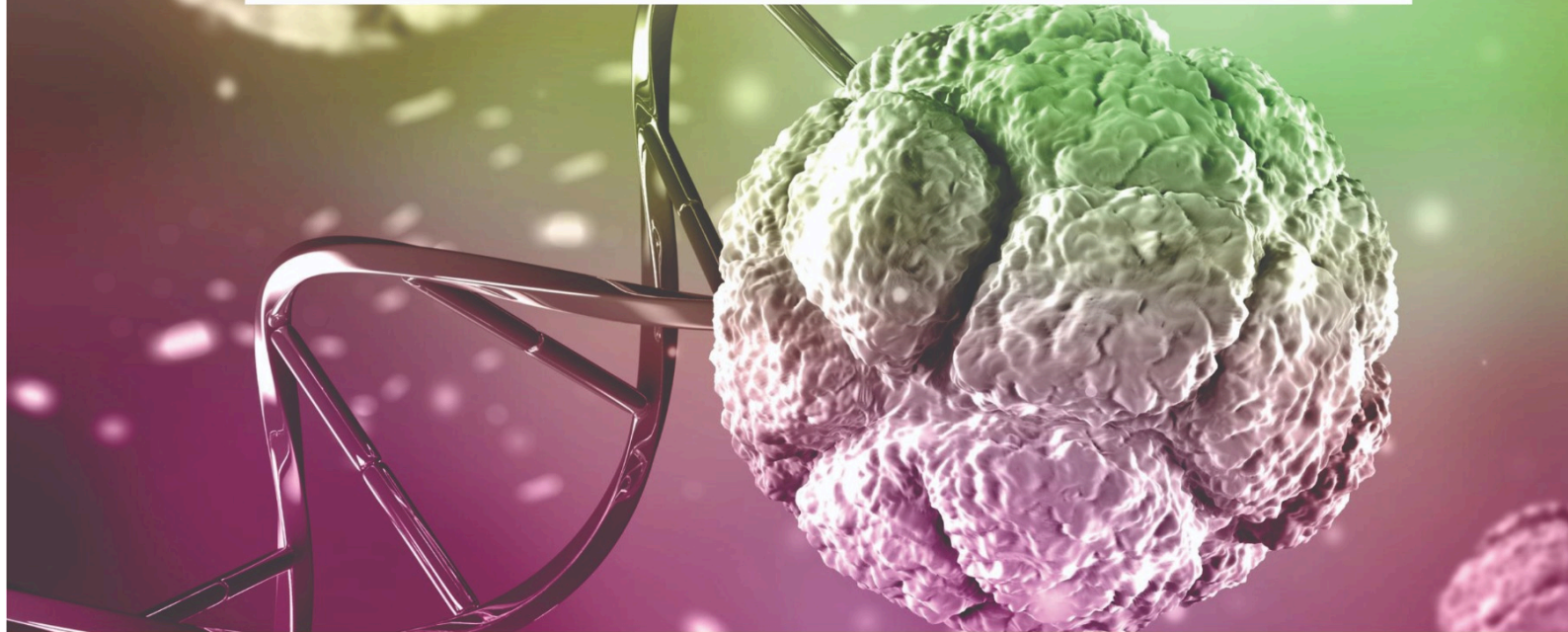
*Prelims Online Batch*

# BIOLOGY

*LECTURE*

7

**Cell Division (Mitosis & meiosis)**



## Cell Division (Mitosis & Meiosis)

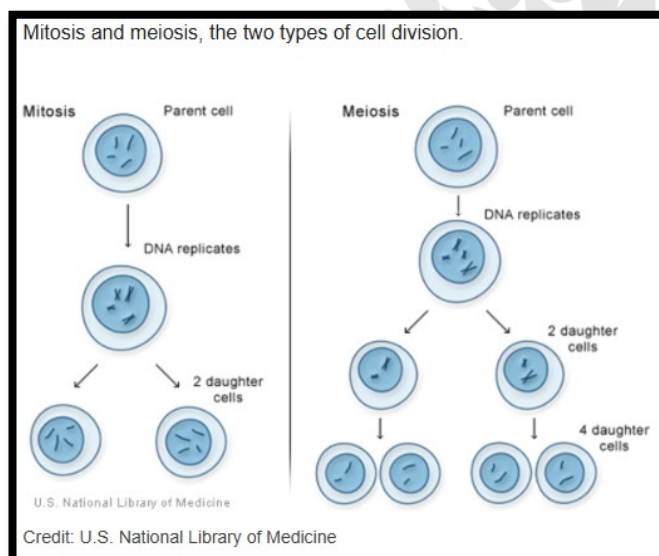
### Cell Division

Cell division is the process by which a parent cell divides into two or more daughter cells. It is essential for growth, development, repair of tissues, and reproduction in living organisms.

Cell division ensures the continuity of genetic information from one generation to another.

There are two main types of cell division:

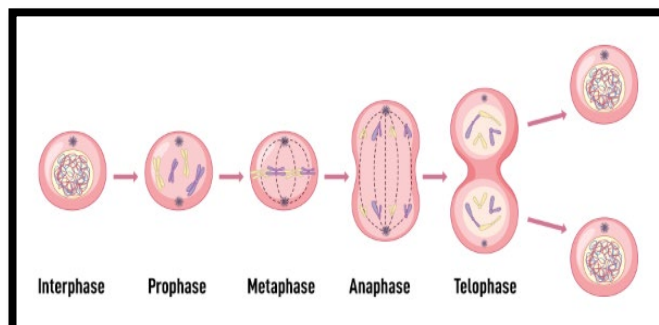
- Mitosis
- Meiosis



| Cell Division       |             |
|---------------------|-------------|
| Mitosis             | Meiosis     |
| Equational division | Reductional |

### Mitosis

- Mitosis is a type of cell division in which one parent cell divides into two genetically identical daughter cells, each having the same number of chromosomes as the parent cell.



### Occurrence

- Mitosis occurs in somatic (body) cells.

Functions of Mitosis

### Growth of organisms

- Repair and replacement of damaged cells
- Asexual reproduction in some organisms
- Maintenance of chromosome number

| Cell Division                                                                |                        |                         |
|------------------------------------------------------------------------------|------------------------|-------------------------|
| Interphase                                                                   | Division phase         |                         |
| Most active phase<br>Cell division<br>Preparation phase<br>for Cell division | Mitosis                | Meiosis                 |
|                                                                              | Equational<br>Division | Reductional<br>Division |

- Mitosis is an equational division occurring in somatic cells, producing two
- genetically identical diploid daughter cells from a parent cell.
- It maintains chromosome numbers, enables growth, tissue repair, and asexual
- reproduction, and consists of karyokinesis (nucleus division) followed by
- cytokinesis (cytoplasm division).
- **Prophase:** Chromatin condenses into chromosomes, centrosomes move to opposite poles, and the nuclear envelope/nucleolus disappears.
- **Metaphase:** Chromosomes align at the metaphase plate (equator), and spindle fibers attach to the kinetochores of sister chromatids.
- **Anaphase:** Centromeres split, sister chromatids separate and are pulled towards opposite poles.
- **Telophase:** Chromosomes decondense at poles, and the nuclear envelope reforms around each daughter nucleus.



**Cytokinesis (Division of Cytoplasm)**

- **Animal Cells:** A cleavage furrow forms in the plasma membrane, deepening inwards to separate the cell.
- **Plant Cells:** A cell plate forms in the center and grows outward to divide the cell, as rigid walls prevent furrowing.

**Significance of Mitosis:**

- **Growth:** Essential for increase in cell number.
- **Repair:** Replaces old, damaged, or dead cells.
- **Genetic Stability:** Ensures daughter cells are identical to the parent.
- **Asexual Reproduction:** Enables reproduction in many organisms

**Meiosis**

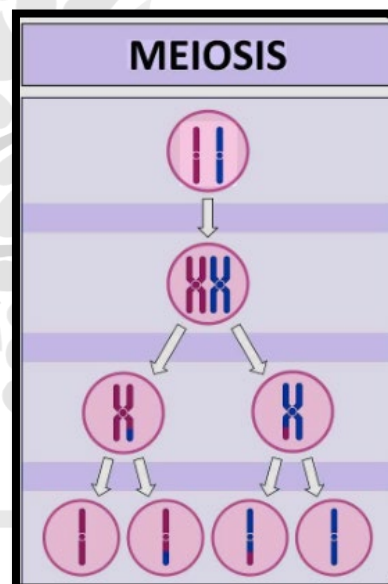
- Meiosis is a reductional division occurring in germ cells, producing four haploid daughter cells from one diploid parent cell via two sequential divisions (Meiosis I & II), with only one DNA replication phase.
- It is essential for producing gametes (gametogenesis), maintaining chromosome number across generations, and increasing genetic variation through crossing over
- **Interphase:** DNA replicates during the S-phase, similar to mitosis, followed by a shorter G<sub>2</sub> phase.
- **Meiosis I (Reductional Division):** Reduces chromosome number to half.

**Prophase I (Longest Phase):**

- **Leptotene :** Chromosomes become visible.
- **Zygotene :** Homologous chromosomes pair up (synapsis) to form bivalents or tetrads.
- **Pachytene :** Crossing over occurs between non-sister chromatids of homologous chromosomes via the enzyme recombinase.
- **Diplotene:** Dissolution of the synaptonemal complex; chiasmata (X-shaped structures) become visible.
- **Diakinesis:** Terminalization of chiasmata; chromosomes fully condense.

- **Metaphase I:** Bivalents align at the equator; microtubules attach to homologous pairs.
- **Anaphase I:** Homologous chromosomes separate, but sister chromatids remain attached.
- **Telophase I:** Nuclear membrane and nucleolus reappear, resulting in two daughter cells (diad).
- **Meiosis II (Equational Division):** Similar to mitosis, it splits the sister chromatids, resulting in four haploid daughter cells.
- **Stages:** Prophase II, Metaphase II, Anaphase II, Telophase II.

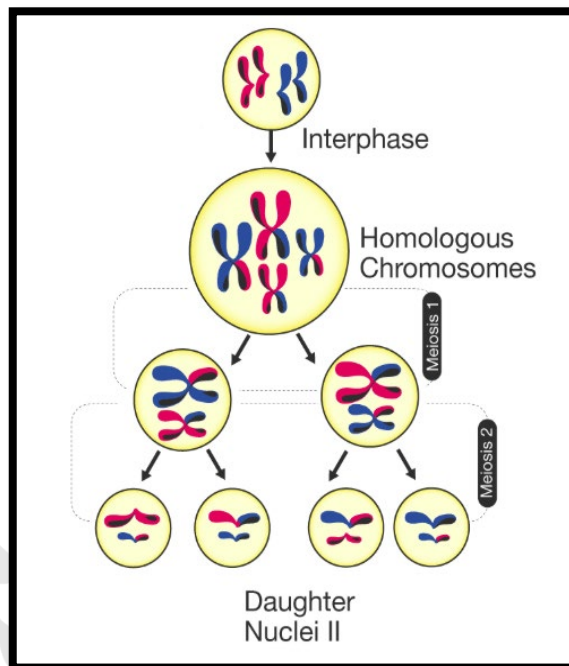
Meiosis is a type of cell division that produces four genetically different daughter cells, each having half the number of chromosomes of the parent cell.



- Meiosis is a reductional division occurring in germ cells, producing four haploid daughter cells from one diploid parent cell via two sequential divisions (Meiosis I & II), with only one DNA replication phase.
- It is essential for producing gametes (gametogenesis), maintaining chromosome number across generations, and increasing genetic variation through crossing over
- **Interphase:** DNA replicates during the S-phase, similar to mitosis, followed by a shorter G<sub>2</sub> phase.
- **Meiosis-I (Reductional Division):** Reduces chromosome number to half.
- **Leptotene:** Chromosomes become visible.



- **Zygotene:** Homologous chromosomes pair up (synapsis) to form bivalents or tetrads. Pachytene: Crossing over occurs between non-sister chromatids of homologous chromosomes via the enzyme recombinase.
- **Diplotene:** Dissolution of the synaptonemal complex; chiasmata (X-shaped structures) become visible.
- **Diakinesis :** Terminalization of chiasmata; chromosomes fully condense.
- **Metaphase I :** Bivalents align at the equator; microtubules attach to homologous pairs.
- **Anaphase I:** Homologous chromosomes separate, but sister chromatids remain attached.
- **Telophase I:** Nuclear membrane and nucleolus reappear, resulting in two daughter cells (diad).
- **Meiosis II (Equational Division):** Similar to mitosis, it splits the sister chromatids, resulting in four haploid daughter cells. ]
- **Stages:** Prophase II, Metaphase II, Anaphase II, Telophase II.
- Metaphase I is a critical, unique stage in the first meiotic division (meiosis I) of eukaryotic cell division.
- Unlike mitosis or metaphase II, metaphase I is defined by the alignment of homologous chromosome pairs (bivalents or tetrads) along the metaphase plate rather than individual chromosomes
- Anaphase I is the stage in meiosis I where homologous chromosomes, rather than sister chromatids, separate and move to opposite poles of the cell.
- Driven by the shortening of kinetochore microtubules, this reductional division separates bivalents (tetrads), ensuring each daughter cell receives one chromosome from each pair.
- Telophase 1 is the final stage of meiosis I, where haploid sets of chromosomes reach opposite poles of the cell, the nuclear envelope reforms, and cytokinesis divides the cytoplasm to create two unique, nonidentical daughter cells.



| Difference Between Mitosis and Meiosis |                |                       |
|----------------------------------------|----------------|-----------------------|
| Feature                                | Mitosis        | Meiosis               |
| Type of Cells                          | Somatic Cells  | Germ cells            |
| Number of divisions                    | One            | Two                   |
| Daughters Cells                        | 2              | 4                     |
| Chromosome number                      | Same as parent | Half of parent        |
| Genetiic similarity                    | Identical      | Genetically different |
| Crossing over                          | Absent         | Present               |